EXPT-08

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NMAP TO DISCOVER LIVE HOSTS USING NMAP SCANS (ACP, ICMP, TCP/UDP).

AIM:-

To discover live hosts in a network using Nmap scans (ARP, ICMP, TCP/UDP) on the TryHackMe platform Room Link: https://tryhackme.com/room/lnmap01.

STEPS AND PROCEDURE:-

1. Open the TryHackMe platform and launch the Nmap Live Host Discovery room.
2. Study the given network topology and identify the target hosts.
3. Observe ARP request and response packets in the local network.
4. Perform ARP host discovery using:

   nmap -PR -sn TARGET.

5. Perform the ICMP protocol to send a Ping request between two systems and observe the corresponding Ping Reply.
6. Observe the packet flow and identify the ARP request and ARP response that may occur before ICMP communication on the same subnet.
7. Perform ICMP ~~Ping~~ Echo scanning using:

   nmap -PE -sn TARGET.

8. Perform ICMP Timestamp scanning using:

nmap -PP TARGET

9. Perform ICMP Address mask scanning using:
   nmap -PM TARGET.

10. Perform TCP SYN Ping scanning using:
    nmap -PS23 TARGET.

11. Perform TCP ACK Ping scanning using:
    nmap -PA80 TARGET.

12. Perform reverse DNS lookup to obtain the hostnames associated with the discovered IP addresses. The -n option can be used when DNS resolution is not required, while -R can request. DNS resolution for all hosts.

13. Perform UDP host discovery and observe the responses recieved from the target. An ICMP port unreachable response can indicate that the target host is active.

14. Record and compare the live hosts discovered using ARP, ICMP, TCP, UDP.

15. Analyze the advantage and limitations of each host discovery method based on the observations obtained from the TryHackme exercises.

OBSERVATION:-

i) ARP:-
   Used to discover live hosts on the same subnet.

ii) ICMP:-
   Uses Echo or Timestamp or address mask requests for host discovery.

iii) TCP:-

Uses SYN or ACK packets to determine whether a host is active.

iv) UDP:-

A response such as ICMP Port unreachable can indicate that the host is active.

RESULT:-

Thus, live hosts were successfully discovered using ARP, ICMP, TCP and UDP using Nmap techniques. The experiment demonstrated how different network protocols can be used for live host discovery.